

Víctor Manuel Chávez Pérez

Professor–Researcher · Applied Physics & Science Education

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National Researcher System · Candidate (SECIHTI) 2025–2029



PROFILE

I am a physicist from the University of Guadalajara and hold a PhD in Applied Physics from the University of Vigo. My research blends atmospheric and climate physics with large-scale data analysis: I study sudden stratospheric warmings (SSW), the Brewer–Dobson circulation and their response under climate-change scenarios using models such as CMIP5 and WACCM.

Beyond basic research, I bring physics to real-world problems: I am developing a flood early-warning system for the Guadalajara Metropolitan Area that integrates weather radar, hydrological runoff modeling over high-resolution LiDAR terrain and remote sensing. My work also spans physical oceanography, remote sensing and applied optics.

I am also interested in how generative AI can transform science education: I design learning assistants and teaching materials that I evaluate with students themselves. Alongside this, I co-founded the Tlakakini Science & Technology Club, devoted to science outreach for children. I believe the best science is the kind that gets shared.

RESEARCH INTERESTS

Sudden stratospheric warmings · Stratospheric dynamics · General circulation models (CMIP5 / WACCM) · Climate change · Physical oceanography & remote sensing · Applied optics · High-performance computing · Science education & outreach · Generative AI in science education · Machine Learning applied to climate & atmospheric data · Student perception of AI-based learning tools

EDUCATION

- 2017 – 2024** **PhD in Applied Physics**
Universidade de Vigo · Spain
Thesis: Sensitivity of sudden stratospheric warmings to simulation and analysis conditions in models.
- 2011 – 2012** **MSc in Climate Science: Meteorology, Physical Oceanography & Climate Change**
Universidade de Vigo · Spain
Temporal and spatial variability of the Minho river plume (MODIS 250 m). Top honors.
- 2009 – 2011** **MSc in Hydrometeorology — Meteorology & Physical Oceanography**
Universidad de Guadalajara · Mexico
Seasonality and anomalies of oceanographic variables in the Gulf of Mexico via remote sensing.
- 2010 – 2011** **Diploma in Teaching Competencies**
Universidad de Guadalajara · Mexico
Pedagogical training for university teaching.
- 2000 – 2008** **BSc in Physics**
Universidad de Guadalajara · Mexico
Thesis: Ocean and coastal meteorology of Nayarit (San Blas, 1999–2002). Grade 95/100.

EXPERIENCE

- 2025 – present** **Lecturer "B"**
Centro Universitario de Tlaquepaque · UdeG · Tlaquepaque, Mexico
Department of Basic Sciences: Physics and Mathematics.
- 2025 – 2026** **Lecturer**
Centro Universitario Bonpland y Humboldt · Online
PhD in Aerospace Engineering: Research Methodology and Fluid Mechanics.
- 2019 – present** **Lecturer "B"**
Universidad Tecnológica de Jalisco · Guadalajara, Mexico
Mechatronics Division — Basic Sciences.
- 2016 – 2024** **External Collaborating Researcher**
Universidade de Vigo · EPhysLab · Ourense, Spain
Doctoral research on sudden stratospheric warmings in CMIP5 and WACCM models; statistical analysis of large climate datasets.

- 2012 – 2016 Predoctoral Researcher (FPI Fellowship)**
 Universidade de Vigo · EPhysLab · Ourense, Spain
 Statistical analysis of large-volume data for climate-change research with general circulation models.
- 2019 – 2020 Faculty Lecturer**
 Tecmilenio · Campus Guadalajara · Guadalajara, Mexico
 High school: Matter & Energy, Integral Calculus, Science & Technology.
- 2019 Physics Instructor**
 UNITEC · Campus Guadalajara · Guadalajara, Mexico
 Engineering programs: Statics.
- 2012 – 2013 Physics & Math Instructor**
 Universidad de Especialidades · Guadalajara, Mexico
 Mechanics, Introduction to Physics and Electromagnetism.
- 2008 – 2011 Professor–Researcher**
 Instituto Tecnológico Superior de Zapopan · Zapopan, Mexico
 Mathematics for engineering; parallel computing applied to oceanography.

PUBLICATIONS

- 2025 Observación Astronómica en México: Pasando de lo Visible a lo Invisible**
 Almaguer-Gómez C., Medina J. F., Chávez-Pérez V. M., et al. · *Óptica Pura y Aplicada* 58(1):15 · DOI: 10.7149/OPA.58.1.51200
- 2025 Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations**
 Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere* 16(5) · DOI: 10.3390/atmos16050628
- 2025 Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios**
 Chávez-Pérez V. M., et al. · *Climate* 13(10) · DOI: 10.3390/cli13100207
- 2023 Estacionalidad y Anomalías del Nivel del Mar en el Caribe por Medio de Datos de Altimetría**
 Palacios Hernández E., Chávez Pérez V. M., Sierra Romero L. L. · *Generis Publishing* · ISBN: 979-8-88676-642-4
- 2022 Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Sudden Stratospheric Warmings**
 Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere* 13(4) · DOI: 10.3390/atmos13040546
- 2018 The climatology of dust events over the European continent using data of the BSC-DREAM8b model**
 Mandija F., et al. · *Atmospheric Research* 209:144–162 · DOI: 10.1016/j.atmosres.2018.03.006
- In preparation**
- 2026 Science Beyond the Classroom: Impact Assessment of a Monthly Science Club During National Non-School Days in Mexico**
 Almaguer-Gómez C., Chávez-Pérez V. M., et al.
- 2026 Student Perception of a Generative AI-Based Calculus Learning Assistant: A Mixed-Methods Study**
 Chávez-Pérez V. M., et al.
- 2026 Assessment of the lightning activity response to aerosol optical depth**
 Mandija F., Chávez-Pérez V. M.

CONFERENCES & TALKS

- 2022 LXV Congreso Nacional de Física**
 2–7 oct 2022 · Zacatecas, Mexico
 • Análisis de la frecuencia de ocurrencia de los calentamientos súbitos estratosféricos en el Hemisferio Sur en los modelos de altura estratosférica del CMIP5
 • Caracterización óptica de resina transparente de impresión 3D
- 2020 LXIII Congreso Nacional de Física**
 4–9 oct 2020 · Online
 • Impacto en las significancias de las características básicas de los SSW al comparar modelos del CMIP5 con datos de reanálisis en periodos pre-satelital y pos-satelital
 • Impacto en la frecuencia de ocurrencia de los calentamientos súbitos estratosféricos principales en distintos escenarios de cambio climático usando CMIP5

- 2018 American Geophysical Union (AGU) Fall Meeting 2018**
10–14 dic 2018 · Washington, USA
• Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Stratospheric Sudden Warmings
- 2018 Reunión Anual de la Unión Geofísica Mexicana**
28 oct–2 nov 2018 · Puerto Vallarta, Mexico
• Cambios en la climatología de los calentamientos súbitos estratosféricos principales usando WACCM con alta resolución vertical
• Evaluación de la capacidad de reproducción de los calentamientos súbitos estratosféricos principales en el conjunto de datos históricos de CMIP5
• Calentamientos súbitos estratosféricos principales bajo escenarios de cambio climático con CMIP5
- 2017 European Geosciences Union (EGU) General Assembly 2017**
23–28 abr 2017 · Vienna, Austria
• Estimation of the variability of dust aerosol optical depth over several European regions based on forecast model data
- 2016 LIX Congreso Nacional de Física (SMF)**
2–7 oct 2016 · León, Mexico
• Teleconexión estratosférica y los calentamientos súbitos estratosféricos
• Patrones característicos en las variables estratosféricas de la Oscilación Cuasi-Bienal
• Anomalías estratosféricas antes y después de los calentamientos súbitos estratosféricos para las distintas fases de la Oscilación Cuasi-Bienal
- 2016 SPARC Workshop SHARP 2016 — Stratospheric Change and its Role for Climate Prediction**
11–15 jul 2016 · Berlin, Germany
• Changes in the stratosphere before and after SSW events in the different phases of the Equatorial QBO
• Differences in the detection and classification of the Stratospheric Sudden Warming over the three reanalyses for the period 1979–2014
- 2015 European Geosciences Union (EGU) General Assembly**
12–17 abr 2015 · Vienna, Austria
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings
- 2015 SPARC Regional Workshop — Role of the stratosphere in climate variability and prediction**
12–13 ene 2015 · Granada, Spain
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings
- 2010 Reunión Anual de la Unión Geofísica Mexicana**
7–12 nov 2010 · Puerto Vallarta, Mexico
• Estacionalidad y anomalías de las variables de percepción remota oceanográficas en el Golfo de México
• Estacionalidad y anomalías de la superficie del nivel del mar en el Mar Caribe por medio de datos de altimetría
• Alternativa para el procesamiento de datos oceanográficos de gran cómputo mediante programación en paralelo con MatLab
- 2010 LIII Congreso Nacional de Física**
25–29 oct 2010 · Boca del Río, Veracruz, Mexico
• Supercómputo aplicado a programación en paralelo para optimizar procesos en Astrofísica y Oceanografía Física
- 2009 LII Congreso Nacional de Física (SMF)**
26–30 oct 2009 · Acapulco, Guerrero, Mexico
• Espectros de frecuencias de datos meteorológicos de Isla María Madre
• Comparación del viento costero y marino de la zona costera de San Blas, Nayarit (1999–2002)

ACTIVE RESEARCH PROJECTS

- 2026 – actual Flood early-warning system for the Guadalajara Metropolitan Area — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Development of an urban flood early-warning system that combines remote sensing (IAM-UdeG weather radar), hydrological runoff modeling over high-resolution LiDAR terrain, and official flood inventories. The system classifies risk by sub-basin in real time and estimates water arrival time, aiming to calibrate thresholds through a retrospective analysis with historical Civil Protection/Fire Department records.
- 2026 - Generative AI for lab practice reports (Google Gems) — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Implementation of Google Gems as a writing and feedback assistant for students to draft and self-assess their lab practice reports in physics courses.

- 2025 – actual** **AI-generated educational comics for science teaching — Principal investigator**
 Universidad de Guadalajara · Funding: No external funding
 Development and evaluation of educational comics created with generative AI tools as a didactic resource for teaching physics, mathematics and science at basic and upper-secondary levels.
- 2025 – actual** **Pedagogical Calculus Assistant with Google Gems: development and student perception — Principal investigator**
 Universidad Tecnológica de Jalisco · Funding: Technological University of Jalisco
 Design and implementation of a Calculus learning assistant based on Google Gems, with quantitative and qualitative study of students' perception, satisfaction and academic performance.
- 2025 – 2028** **Optical elements to enhance visual acuity via wavefront coding — Collaborator**
 Universidad de Guadalajara · Funding: Universidad de Guadalajara
 Design and validation of optical elements that improve visual acuity through wavefront coding.
- 2025 – actual** **World Wide Lightning Location Network (WWLLN) — Collaborator · antenna lead**
 University of Washington · Funding: University of Washington
 Global lightning location network. In charge of the receiving antenna/node and real-time data delivery.
- 2024 – actual** **Tlakakini Science & Technology Club — Coordinator**
 CUCEI · Universidad de Guadalajara · Funding: University of Guadalajara · CUCEI
 Science-outreach club for children, with monthly experiment and workshop sessions.

TECH DEVELOPMENTS

Flood early-warning system · Guadalajara Metro Area

Python · FastAPI · Leaflet · NumPy · pysheds · rasterio · GDAL · Docker

Real-time web platform that anticipates urban flooding in the Guadalajara Metropolitan Area. It fuses the IAM-UdeG weather radar (one scan per minute) with a hydrological runoff model built on 5 m LiDAR and Copernicus DEM terrain, plus the IMEPLAN official flood inventory (363 recurrent sites). It produces a per-zone risk light —calibrated against official water-depth records— and a continuous risk grid that routes rainfall downslope, with concentration time and a "water incoming" ETA. Includes live and historical-replay views. Deployed in production.

TEACHING

Centro Universitario Bonpland y Humboldt · Online — Fluid Mechanics, Research Methodology

Centro Universitario de Tlaquepaque · UdeG · Tlaquepaque, Mexico — Physics & Mathematics, Programming Principles, Data Science

Universidad Tecnológica de Jalisco · Guadalajara, Mexico — Physics & Mathematics, Programming Principles, Data Science

Tecmilenio · Campus Guadalajara · Guadalajara, Mexico — Matter & Energy, Integral Calculus, Science & Technology

UNITEC · Campus Guadalajara · Guadalajara, Mexico — Statics

Universidad de Especialidades · Guadalajara, Mexico — Mechanics, Intro to Physics, Electromagnetism

CETI Colomos · Guadalajara, Mexico — Dynamics, Differential Calculus

SCIENCE OUTREACH

- 2024 –** **Tlakakini Science & Technology Club**
 Co-founder. Monthly outreach sessions for children on national non-school days.
- 2009 –** **Physics conferences (AGU, EGU, SMF)**
 Talks at national and international conferences on climate and the stratosphere.
- 2005 –** **Training science communicators**
 Optics and electromagnetism workshops; National Outreach Meeting.
- 2009 –** **Environment & Sustainability Thematic Network**
 Member of the CONACYT environment and sustainability network.
- 2011 –** **Environmental Physics Laboratory (EPHysLab)**
 External collaborating researcher with the University of Vigo group.

GRANTS & HONORS

- 2025–2029** **National Researcher System · Candidate**
 SECIHTI · Mexico
- 2012–2016** **FPI Grant · Predoctoral Researcher**
 Ministerio de Educación · Spain
- 2011–2012** **Climate Science Master's Scholarship**
 Campus do Mar · Spain

2011–2012 **«Ourense Exterior» mobility grant**

Diputación de Ourense · Spain

2009–2011 **Hydrometeorology Master's Scholarship**

CONACYT · Mexico

2009 **Admission to the State Research System**

COECYTJAL · Jalisco, Mexico

Certifications

2025 AI for Educators — Transform your teaching with ChatGPT — Red de Universidades Anáhuac

2025 What is Data Science? — IBM

TOOLBOX

Python · Fortran · MatLab · IDL · NCL · LaTeX · COMSOL · LabVIEW · MPI · Linux · Moodle · Pandas · Scikit-learn · Jupyter